

Computer Memory Hierarchy

Lesson 07 — Speed, Cost and Capacity Trade-offs

JKSSB Computer Awareness

October 8, 2026

Why a Memory Hierarchy Exists

- ▶ A running computer keeps data and instructions in several different stores, not in one single place.
- ▶ **No single memory** is at once fast, large, and cheap.
- ▶ Faster memory is smaller and more expensive per bit; slower memory is larger and cheaper per bit.
- ▶ The **memory hierarchy** arranges these stores from fastest and smallest down to slowest and largest.

The one idea

The hierarchy balances three competing needs: speed, cost, and capacity.

The Ordered Hierarchy

Level	Speed	Typical role
Registers	Fastest, smallest	Values in use right now
Cache (L1/L2/L3)	Very fast	Recently and frequently used data
Main memory (RAM)	Moderately fast	Active programs and data
Secondary (SSD/HDD)	Slower	Permanent file storage
Tertiary/offline (tape, cloud)	Slowest	Backup and archive

Registers are fastest; the order runs down to slowest offline storage.

Registers: Fastest and Smallest

- ▶ Registers sit **inside the CPU**.
- ▶ They are the **fastest** storage in the whole computer.
- ▶ Their capacity is **very small**, so they hold only the values needed at this instant.
- ▶ They hold instructions, addresses, data, and intermediate results while work is in progress.

Speed order

Registers are fastest, then cache, then main memory (RAM), then secondary storage.

Cache Memory

- ▶ **Cache** is a small, very fast memory between the CPU and main memory.
- ▶ It holds **recently and frequently used** data and instructions.
- ▶ Cache is **faster** than main memory (RAM).
- ▶ It keeps copies of data that really live in RAM; it is **not** a backup.

Why cache helps

By keeping the most likely-needed data close to the CPU, cache reduces the average time taken to fetch data.

Cache Levels: L1, L2, L3

Level	Size / speed	Position
L1	Smallest and fastest	Closest to the CPU core
L2	Larger, a little slower than L1	Per CPU core
L3	Largest, slowest of the three (still faster than RAM)	Often shared across cores

Exam depth: L1 is fastest and smallest; L3 is the largest and slowest cache level, but still faster than main memory.

Cache vs Main Memory (RAM)

Point	Cache	Main memory (RAM)
Speed	Faster	Slower
Size	Smaller	Larger
Holds	Recent/frequent data	Active programs and data

Trap alert

Cache is **faster** than main memory, not slower. Reversing this is a common wrong answer.

Main Memory (RAM)

- ▶ Main memory is the computer's **primary memory**.
- ▶ The programs and data in use are held here while the machine runs.
- ▶ **RAM** is **volatile**: its contents are lost when the power is switched off.
- ▶ RAM is larger than cache but slower; the CPU works directly with main memory.

Trap alert

RAM is main memory, not permanent file storage. Switch the machine off and its contents are gone.

ROM: Non-Volatile Primary Memory

- ▶ **ROM** stands for Read-Only Memory.
- ▶ It is **non-volatile**: it keeps its contents without power.
- ▶ In normal operation it is read-only; writing to it needs a special process.
- ▶ It stores **firmware** such as the startup (BIOS) instructions.

Keep the pair straight

RAM is volatile; ROM is non-volatile. Do not swap them.

Primary Memory vs Secondary Storage

Primary memory

- ▶ Directly accessible by the CPU
- ▶ RAM volatile; ROM non-volatile
- ▶ Holds active programs and data

Secondary storage

- ▶ Non-volatile; keeps files without power
- ▶ Larger and cheaper per bit, but slower
- ▶ Reached via main memory and the OS

The CPU does not access secondary storage directly.

Secondary and Tertiary Storage

- ▶ **Secondary storage**: SSD (flash, no moving parts) and HDD (magnetic disks).
- ▶ It is **non-volatile** and holds files and programs between sessions.
- ▶ **Tertiary / offline** storage: magnetic tape, archival media, and cloud backup.
- ▶ It is the **slowest and cheapest** level, used for backup and long-term archive.

Locality of Reference

- ▶ Programs tend to **reuse** the same data and instructions.
- ▶ **Temporal locality**: the same item is accessed again soon after it was last used.
- ▶ **Spatial locality**: items stored at nearby addresses are accessed next.
- ▶ This re-use is exactly why cache works.

Office desktop example

While a spreadsheet is edited, the same active cells and nearby rows are used again and again — so cache keeps that data ready.

Trap Alert: Memory Facts to Keep Straight

- ▶ RAM is **volatile**; ROM is **non-volatile**. (TRAP-01)
- ▶ Cache is **not** a backup; it holds copies of data that live in RAM.
- ▶ RAM is main memory, **not** storage for files.
- ▶ Cache is **faster** than main memory. (PYQ-03)
- ▶ Order of speed: registers → cache → RAM → secondary storage.

Lesson 07: Recall

- ▶ The hierarchy runs registers, cache, main memory (RAM), secondary storage, then tertiary/offline storage.
- ▶ Registers are fastest and smallest; slower levels are larger and cheaper per bit.
- ▶ Cache levels: L1 smallest/fastest, L2 next, L3 largest/slowest (still faster than RAM).
- ▶ Cache is faster than main memory and stores recently used data; it is not a backup.
- ▶ RAM is volatile primary memory; ROM is non-volatile; secondary storage keeps files without power.
- ▶ Locality of reference — temporal and spatial — is why cache helps.